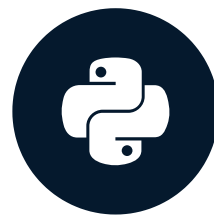


Text-to-query agents

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH



Apoorva Joshi

Senior AI/ML Developer Advocate,
MongoDB

Meet your instructor...

- *Senior AI Developer Advocate, MongoDB*
- 6+ years of Applied ML at MongoDB and Elastic



Coming up...



- World-leading document database

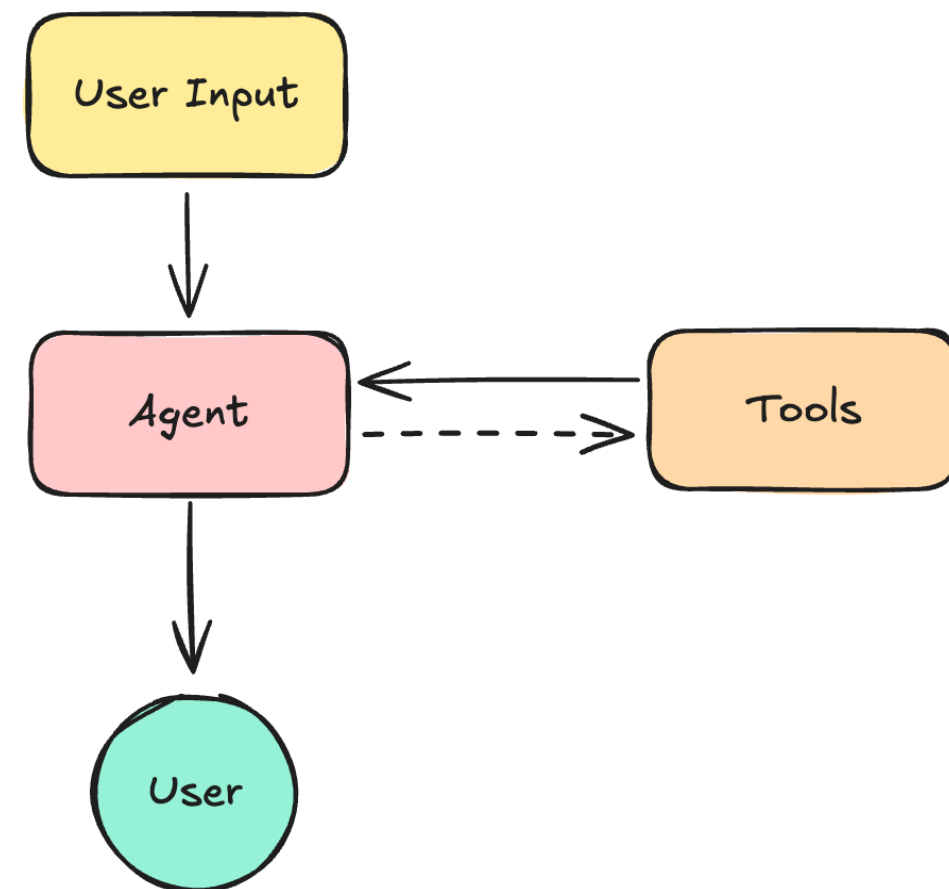
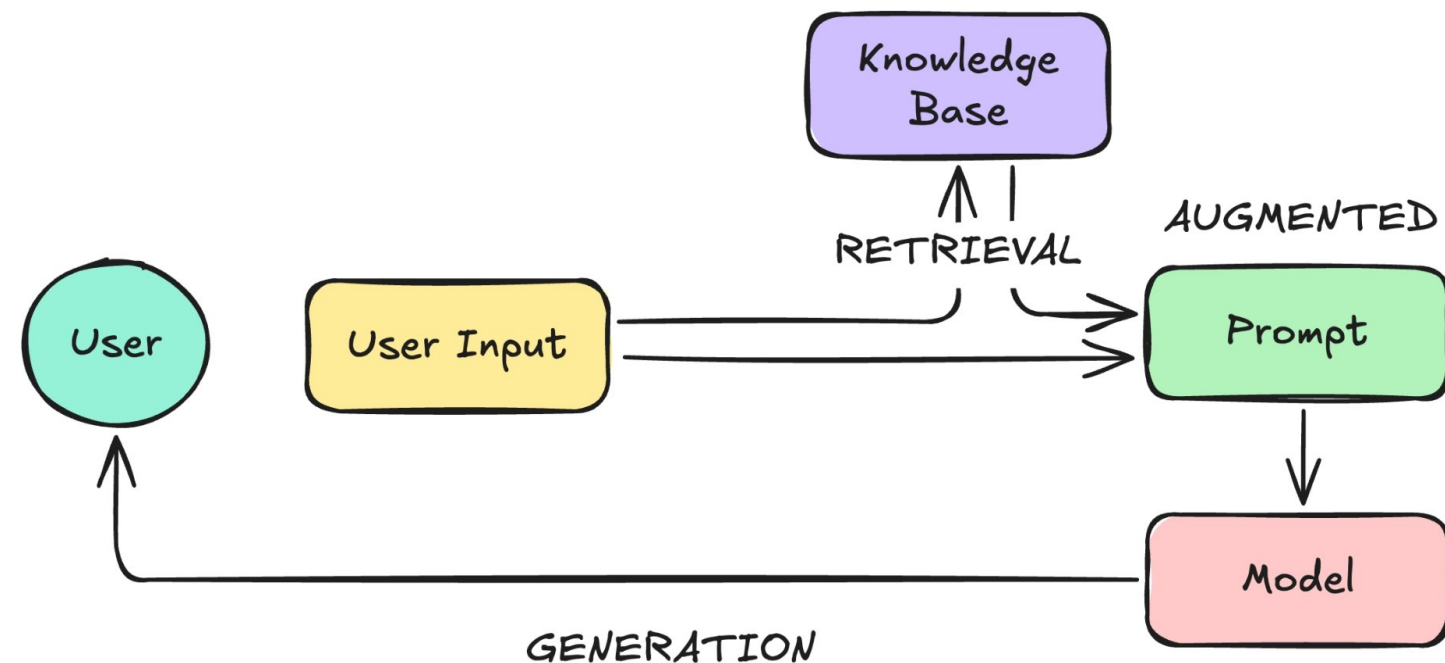
¹ <https://db-engines.com/en/ranking>

RAG and Agents



¹ Generated with GPT-4o

RAG and Agents



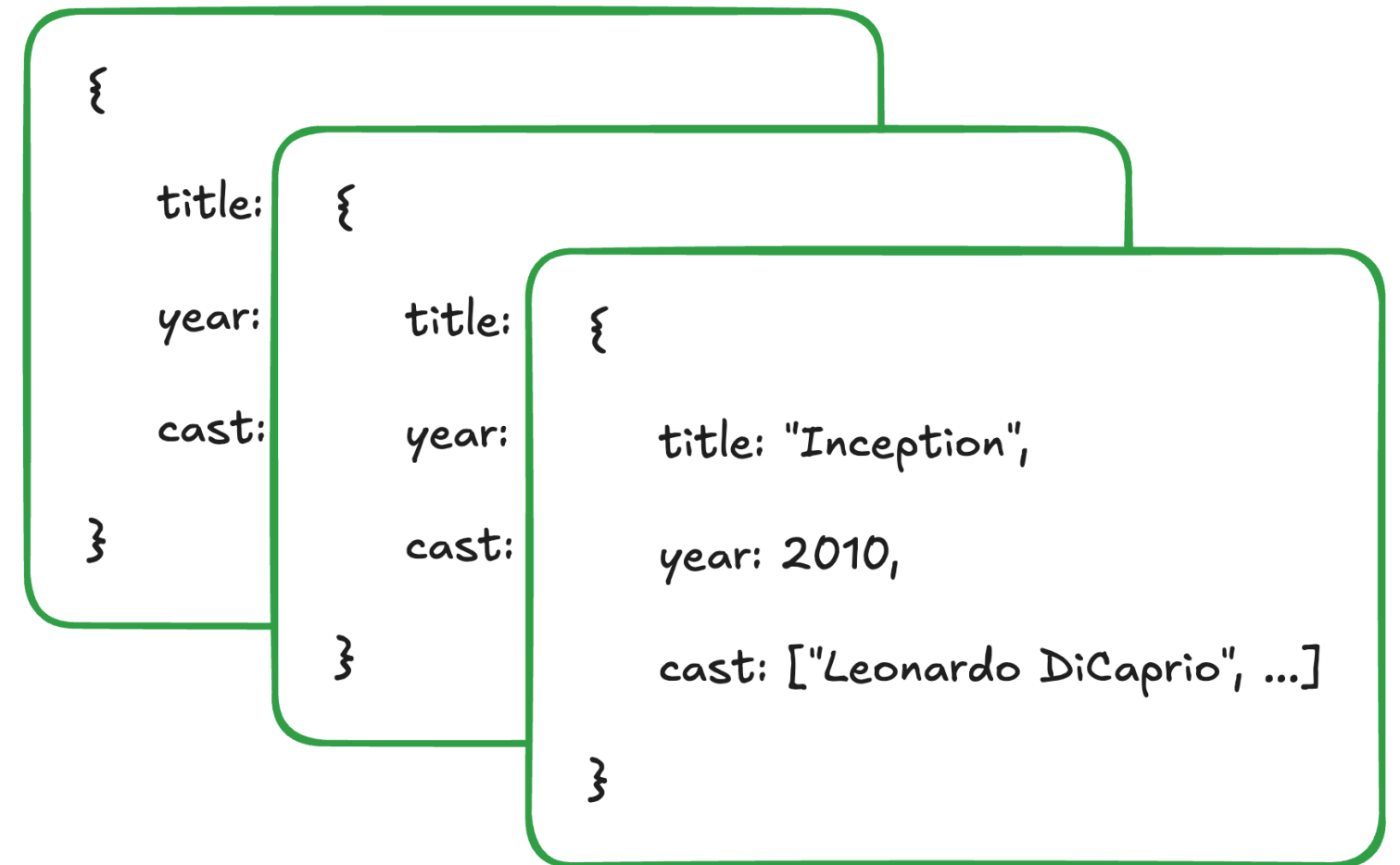
MongoDB Atlas. Fully managed MongoDB in the cloud.

Get ready for higher performance, reduced costs, and greater ease of use.

MongoDB Atlas

- **Cluster:** machinery that supports and runs the database

Goal → build a *text-to-query* agent to answer questions about a database



Video

```
Create a LangChain-compatible MongoDB database instance using your MONGODB_URI and specifying the "sample_mflix" database.

from langchain_mongodb.agent_toolkit.database import MongoDBDatabase

# Access the sample_mflix database using the MongoDBDatabase class
db = MongoDBDatabase.from_connection_string(connection_string=MONGODB_URI, database="sample_mflix")

Initialize the MongoDB database toolkit and view the tools present in the toolkit using the db and llm defined previously.

from langchain_mongodb.agent_toolkit.toolkit import MongoDBDatabaseToolkit
from langchain_openai import ChatOpenAI

llm = ChatOpenAI(
    model="gpt-4o-mini",
    temperature=0
)

# Initialize the MongoDB database toolkit
toolkit = MongoDBDatabaseToolkit(db=db, llm=llm)

# Extract the tools from the toolkit
tools = toolkit.get_tools()
```

Instructions

100XP

- Create a LangChain-compatible MongoDB database instance using your `MONGODB_URI` and specifying the `"sample_mflix"` database.
- Initialize the MongoDB database toolkit and view the tools present in the toolkit using the `db` and `llm` defined previously.
- Test out the four tools from the toolkit with the inputs provided.

Take Hint (-30 XP)

File Environment Saving...

Let's access the `sample_mflix` database using this class and use the resulting database object to initialize the MongoDB database toolkit.

We will use the `from_connection_string` method of the class to connect to your MongoDB cluster using the connection string you obtained previously.

Create a LangChain-compatible MongoDB database instance using your `MONGODB_URI` and specifying the `"sample_mflix"` database.

```
from langchain_mongodb.agent_toolkit.database import MongoDBDatabase

# Access the sample_mflix database using the MongoDBDatabase class
db = _____._____(connection_string=MONGODB_URI, database="_____")
```

Initialize the MongoDB database toolkit and view the tools present in the toolkit using the `db` and `llm` defined previously.

```
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llm = ChatOpenAI(
    model="gpt-4o-mini",
    temperature=0
)
```



Run All

Submit Answer

Video

(mongodb_query)

MongoDB Database Toolkit

```
install pymongo and langchain-mongodb, and assign MONGODB_URI.

let langchain-openai==0.3.28 pymongo==4.13.2 langchain-mongodb==0.6.2

from langchain_openai import ChatOpenAI
from langchain_mongodb.agent_toolkit.database import MongoDBDatabase

llm = ChatOpenAI(
    model="gpt-4o-mini",
    temperature=0
)

db = MongoDBDatabase(connection_string=MONGODB_URI, database="sample_mflix")

toolkit = MongoDBDatabaseToolkit(db=db, llm=llm)

tools = toolkit.get_tools()

print(tools)
```

100XP

- Create a LangChain-compatible MongoDB database instance using your `MONGODB_URI` and specifying the `"sample_mflix"` database.
- Initialize the MongoDB database toolkit and view the tools present in the toolkit using the `db` and `llm` defined previously.
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```

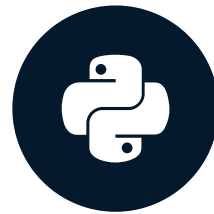
Run All Submit Answer

Let's practice!

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH

Get set up for the hands-on exercises

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH



Apoorva Joshi

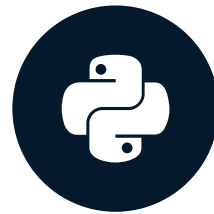
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MongoDB

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TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH

The text-to-query workflow

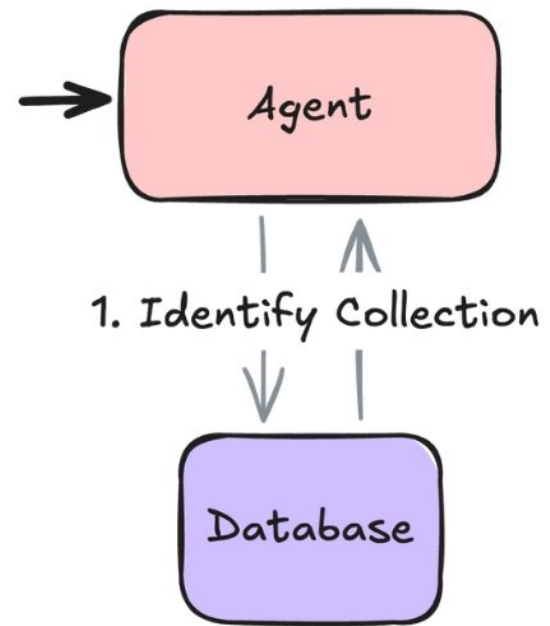
TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH



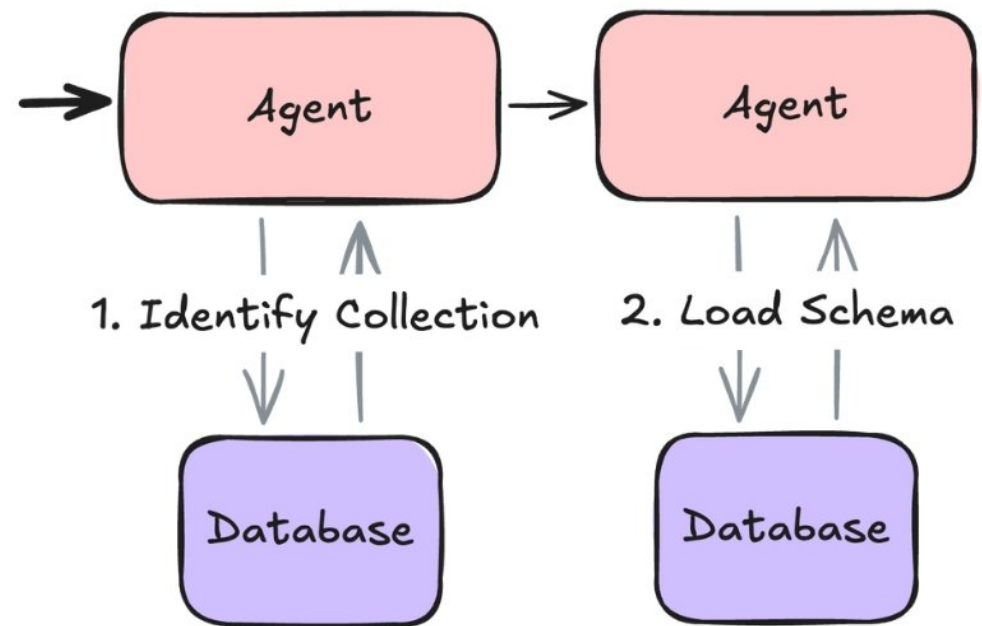
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MongoDB

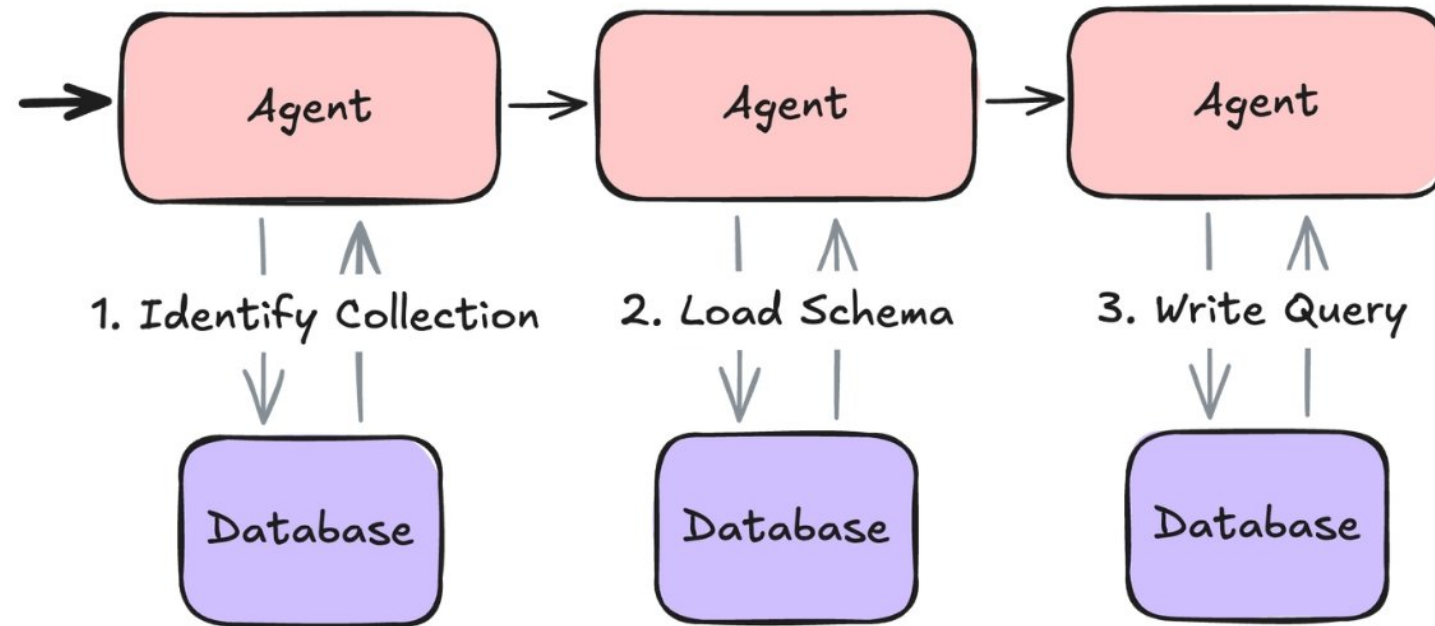
Text-to-query



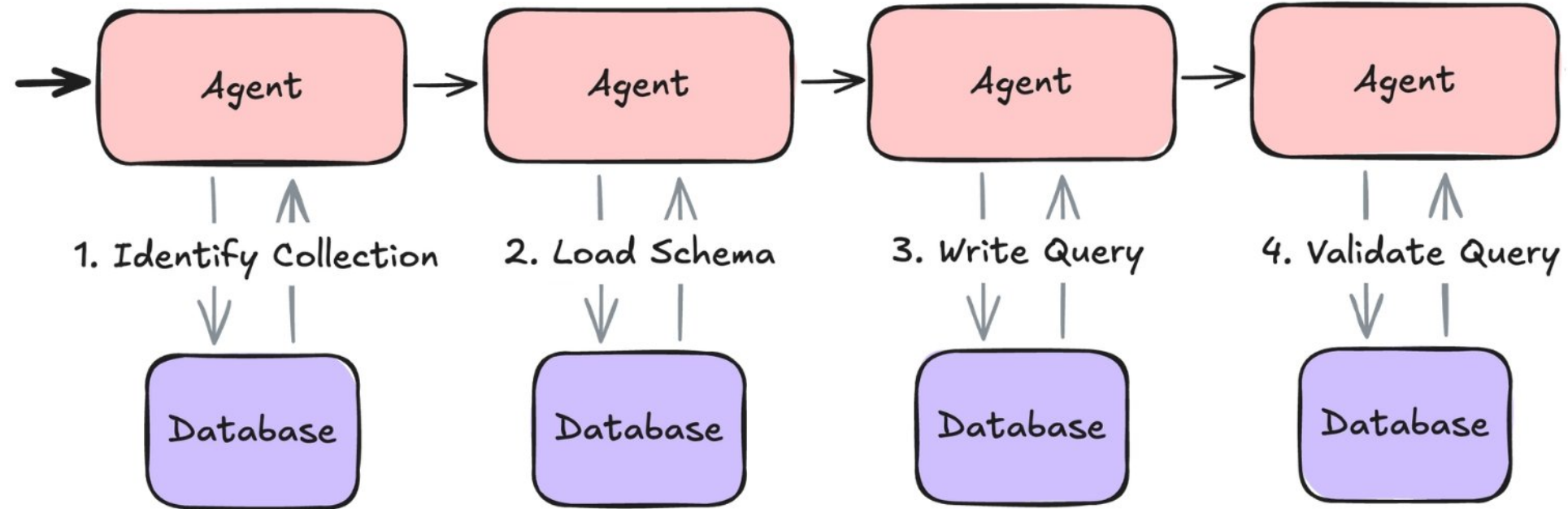
Text-to-query



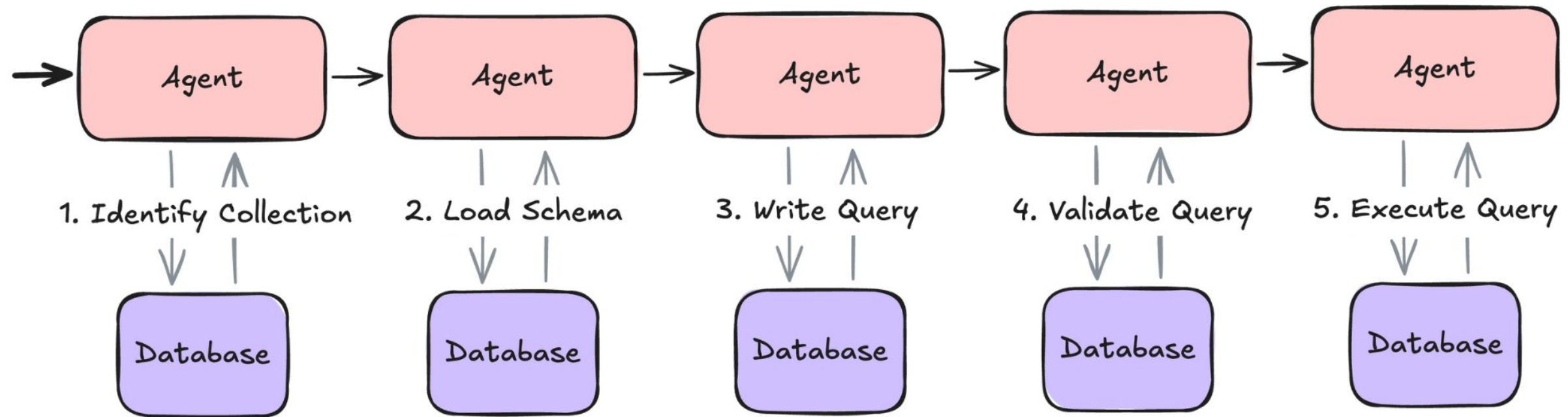
Text-to-query



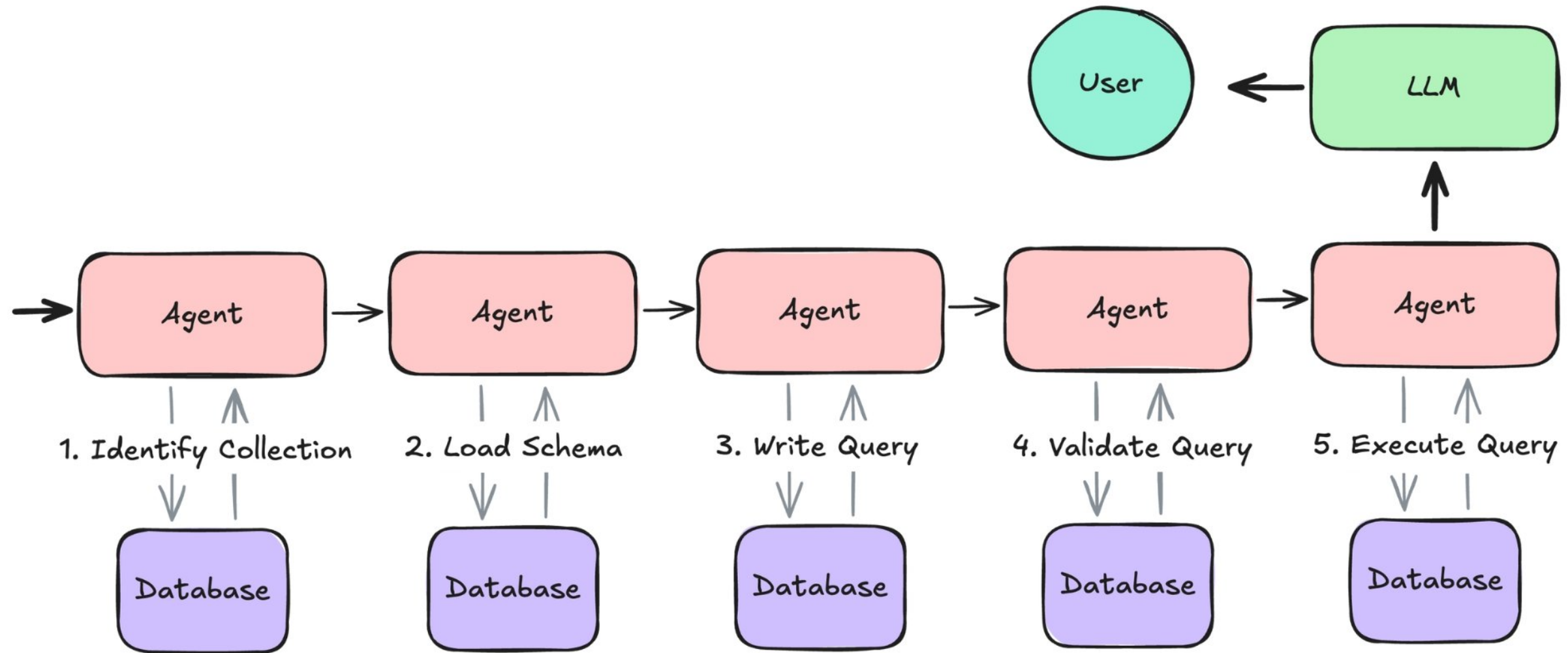
Text-to-query



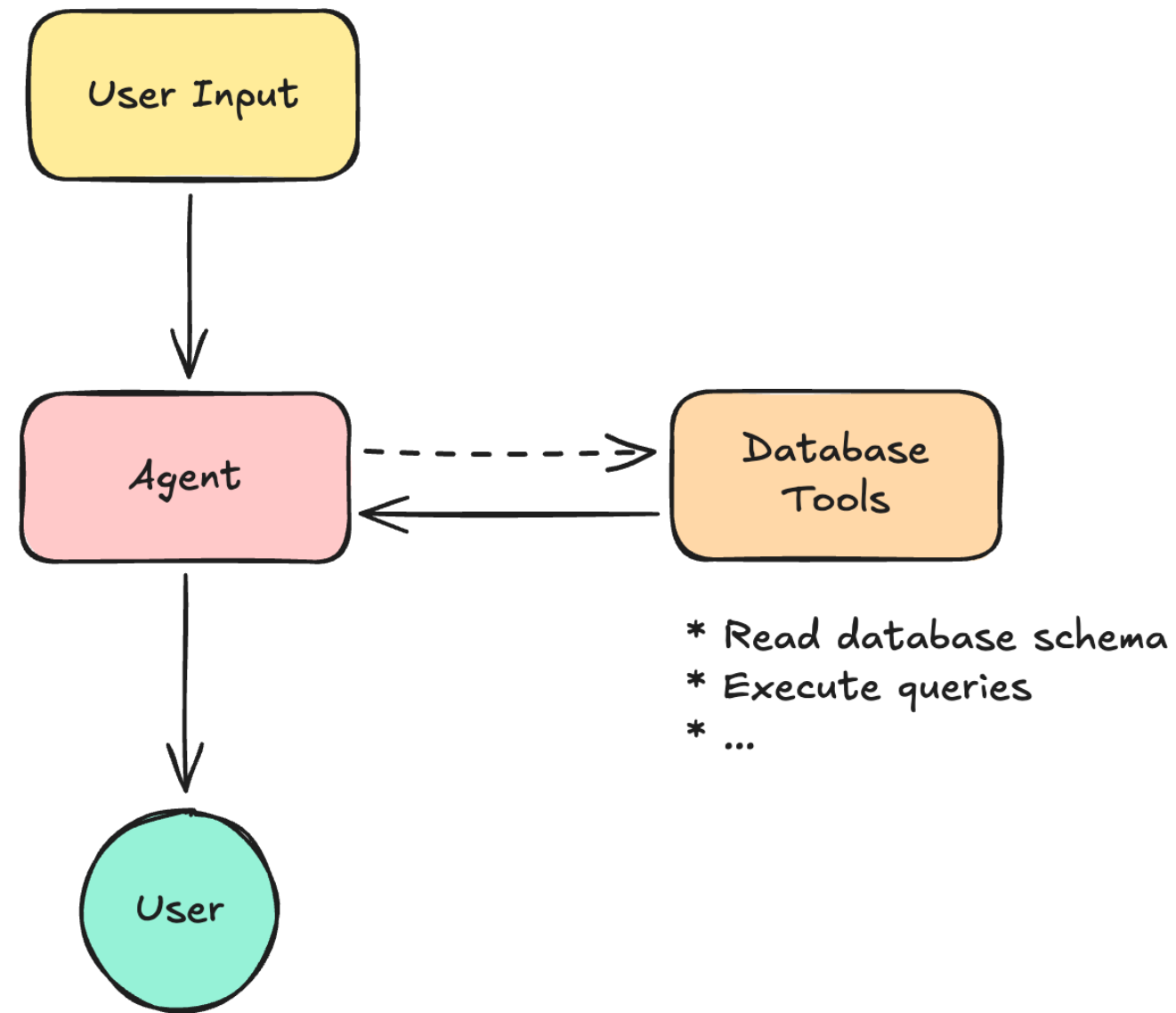
Text-to-query



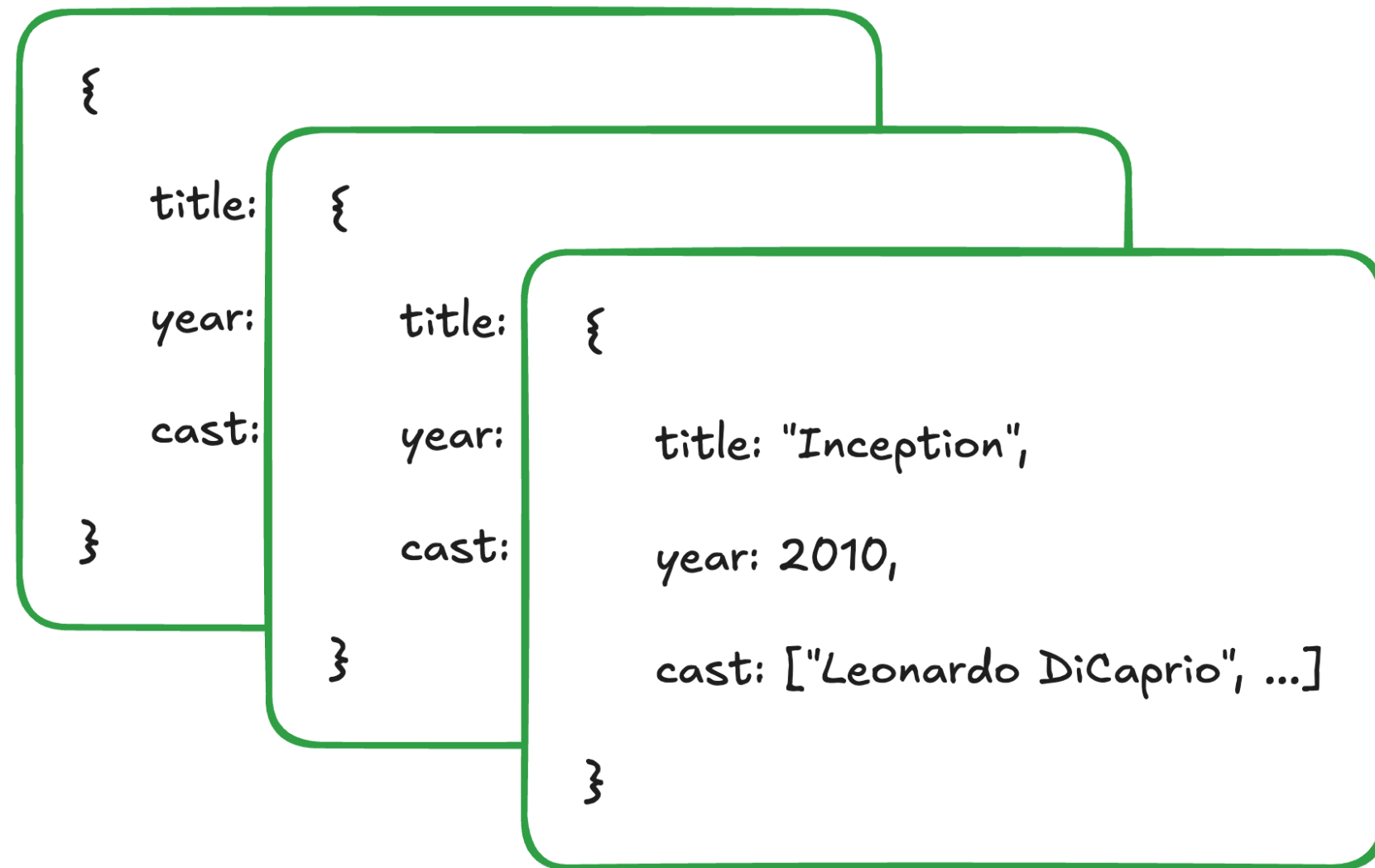
Text-to-query



Text-to-query: the agentic approach



Data: the MongoDB way!



Structured Query Language (SQL)

Example SQL Query

```
SELECT *  
FROM table  
WHERE year >= 1950  
LIMIT 5
```

MongoDB Query API

Example JSON-like Input

```
{ "$and": [  
    {"year": {"$gte": 1990}},  
    {"year": {"$lt": 2000}}  
]
```

MongoDB Query Methods

- `.find()` , `.insert()` , `.update()` ,
 `.delete()`

Example: movies collection

_id	Title	Release Year	Cast	Genres
573a1396f29313caabce3d17	Larks on a String	1990	[Rudolf Hrusènskè, Vlastimil Brodskè]	[Comedy, Drama, Romance]
573a1398f29313caabceab72	The Witching of Ben Wagner	1990	[Sam Bottoms, Harriet Hall, Bettina Rae, Justin Gocke]	[Family]
...	...	...	[...]	[...]

Single filters

```
query = {"genre": "Romance"}
```

Single filters

```
query = {"genre": "Romance"}
```

_id	Title	Release Year	Cast	Genres
573a1390f29313caabcd6377	Wild and Woolly	1917	[Douglas Fairbanks, Eileen Percy, Calvert Carter, Charles Stevens]	[Comedy, Western, Romance]
573a1391f29313caabcd70b4	The Four Horsemen of the Apocalypse	1921	[Pomeroy Cannon, Josef Swickard, Bridgetta Clark, Rudolph Valentino]	[Drama, Romance, War]

Multiple filters

```
query = { "$and": [  
    {"year": {"$gte": 1990}},  
    {"year": {"$lt": 2000}}  
]
```

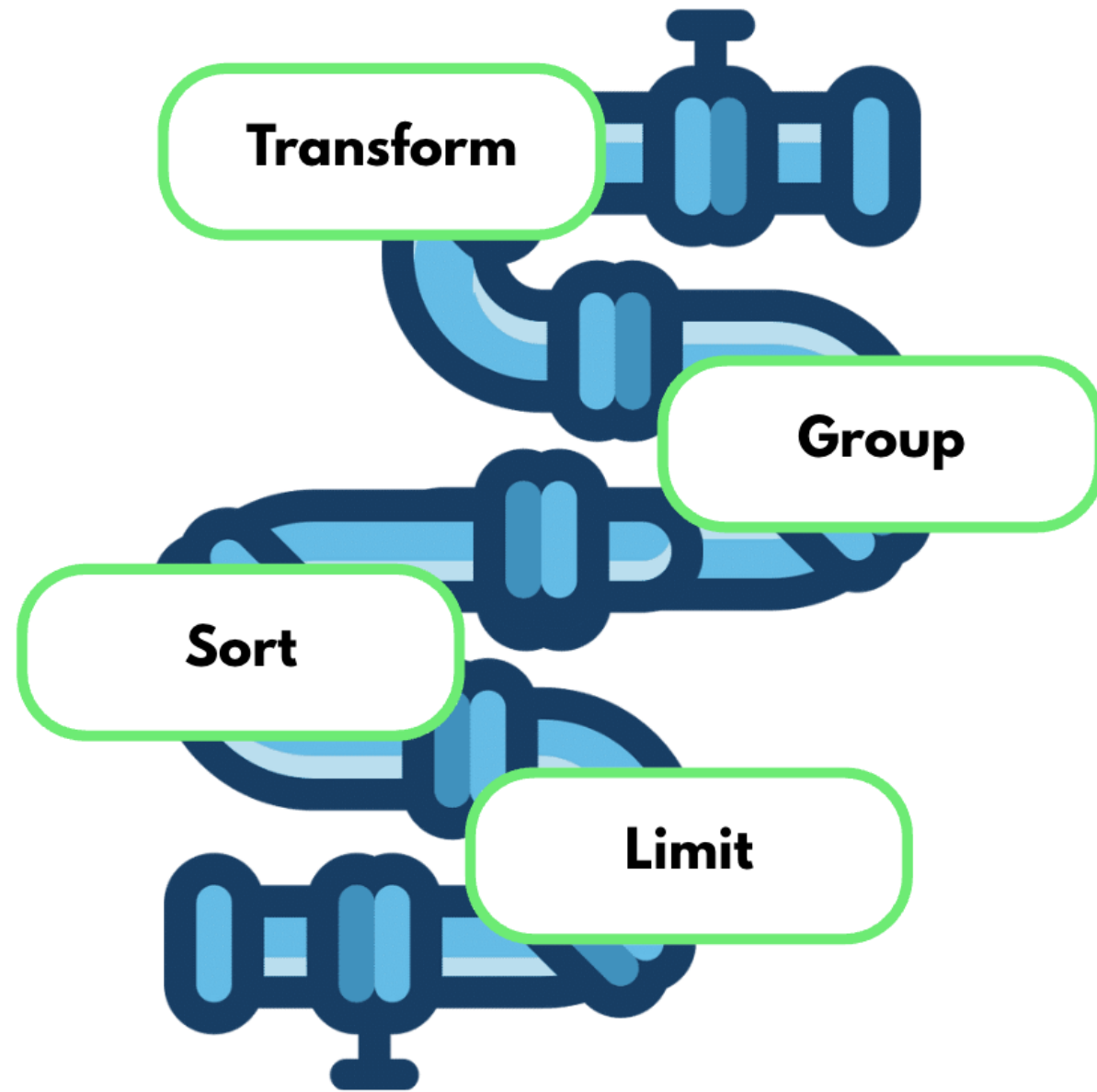
```
}
```

Multiple filters

```
query = { "$and": [
    {"year": {"$gte": 1990}},
    {"year": {"$lt": 2000}}
]
```

_id	Title	Release Year	Cast	Genres
573a1396f29313caabce3d17	Larks on a String	1990	[Rudolf Hrusènskè, Vlastimil Brodskè]	[Comedy, Drama, Romance]
573a1398f29313caabceab72	The Witching of Ben Wagner	1990	[Sam Bottoms, Harriet Hall, Bettina Rae, Justin Gocke]	[Family]

Aggregation pipelines



```
db.<collection>.aggregate([  
    {stage-1},  
    {stage-2},  
    {stage-3},  
    {stage-4},  
])
```

Multiple operations

```
query = [
```

Multiple operations

```
query = [  
    { "$sort": { "released": -1 } },  
  
]
```

- Sort by `released`

Multiple operations

```
query = [  
  { "$sort": { "released": -1 } },  
  { "$limit": 5 },  
  
]
```

- Sort by `released`
- Limit to `5` results

Multiple operations

```
query = [  
  { "$sort": { "released": -1 } },  
  { "$limit": 5 },  
  { "$project": { "title": 1,  
                  "_id": 0 } }  
]
```

- Sort by `released`
- Limit to `5` results
- Project `"title"` column

Multiple operations

```
query = [  
  { "$sort": { "released": -1 } },  
  { "$limit": 5 },  
  { "$project": { "title": 1,  
                  "_id": 0 } }  
]
```

- Sort by `released`
- Limit to `5` results
- Project `"title"` column

Title
The Treasure
Knight of Cups
Sand Castles
Shut In
Dègradè

Grouped aggregations

_id	Title	Release Year	Cast	Genres
573a1396f29313caabce3d17	Larks on a String	1990	[Rudolf Hrusènskè, Vlastimil Brodskè]	[Comedy, Drama, Romance]
573a1398f29313caabceab72	The Witching of Ben Wagner	1990	[Sam Bottoms, Harriet Hall, Bettina Rae, Justin Gocke]	[Family]
...	...	...	[...]	[...]

Grouped aggregations

```
query = [  
    { "$unwind": "$genres" },  
    { "$group": {  
        "_id": "$genres",  
        "numMovies": { "$sum": 1 }  
    }},  
    { "$match": { "numMovies": { "$gte": 50 } } }  
]
```

Grouped aggregations

Genre	Count
Drama	12385
Comedy	6532
Romance	3318
Crime	2457
Thriller	2454
...	...

Pipeline Stages

- `$match` , `$group` , `$facet` , `$geoNear` , `$lookup` , `$merge` , `$search` , `$sort` ...

Operators

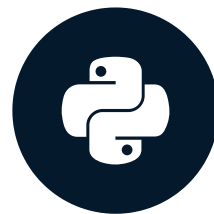
- *Comparison*: `$eq` , `$gt` , `$gte` , ...
- *Math*: `$add` , `$multiply` , `$divide` ...
- *Array*: `$push` , `$reduce` , ...
- *String*: `$toUpper` , `$toLower` , ...
- *Temporal*: `$dateAdd` , `$dateDiff` , ...

Let's practice!

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH

Writing MongoDB queries

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH



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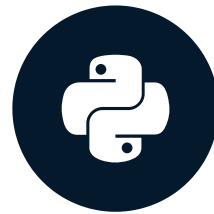
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MongoDB

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More complex aggregations

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH



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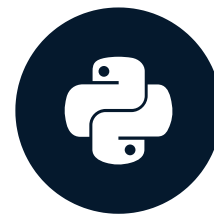
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MongoDB

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TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH

A text-to-query workflow in practice

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH



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Senior AI/ML Developer Advocate,
MongoDB

Let's practice!

TEXT-TO-QUERY AGENTS WITH MONGODB AND LANGGRAPH